

Precision of estimates

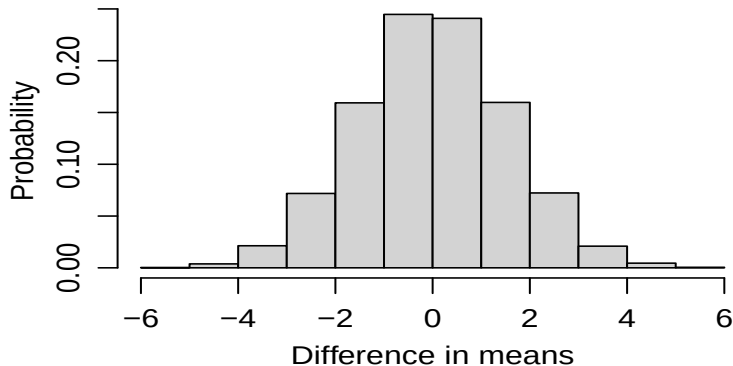
Presentation Outline

- 1 **Review of standard errors**
 - Connection to the sampling distribution
 - Standard error
- 2 **Less familiar aspects using standard errors**
 - Should you report sd or se?
 - Reporting results
- 3 **Important points:**

Standard error

Sampling distribution: reminder

- What could happen when study repeated many times



Standard error - 2

- That distribution has a spread
- Commonly quantified by its standard deviation
- Don't have to construct the sampling distribution
- Math:
 - Data are a simple random sample of n observations from a population with a known sd of σ
 - The sd of the sampling distribution is $\sigma/\sqrt{n}$
- Not useful unless you know σ
- Have an estimate of σ : the sample sd of the observations, usually denoted s
- The estimated sd of the sampling distribution is then $s/\sqrt{n}$

Standard error - 3

- Vocabulary is inconsistent and hence confusing
 - e.g., “standard deviation of the mean” (se computed using σ)
- Notice that $s/\sqrt{n}$ combines two concepts
 - Variability of estimates (spread of the sampling distribution)
 - Whether computed from σ or estimated from the sample sd, s
- I focus on two of the four combinations of concepts
 - Standard deviation (sd): sample (s) or population (σ) *variability in the observations*
 - Standard error (se): estimated *precision of statistic*
- Many different standard errors
 - for a mean, for a difference of means, for a regression slope, for a log odds ratio
 - and for sampling schemes that are not simple random samples or are from finite populations

Should you report sd or se?

What do want to describe?

- Variability in the observations: use sd
- Precision of an estimate: use se

Should you report sd or se?

And correctly state which you are presenting

- Not always done correctly
- From the editor's note on the Med diet retraction:
 - Reviewed 934 papers published in NEJM
 - found 11 with curious results
 - 5 of those because se was incorrectly reported as sd or vice-versa

How many digits should you report?

When reporting an average, should you report 106.4 or 106.386719?

- Computer says sample mean is 106.386719.
- Please don't report that many digits
 - Overstates the precision
 - Implies you know the value is not 106.386718 or 106.386720
 - Clutters the table and makes it harder to see patterns
- Kelley's rule¹
- Report a statistic, e.g., an average, to the leading digit of $se/3$

¹Kelley, 1924, Science 60:524

How many digits should you report? - 2

Example of using Kelley's rule

- Computer reports 106.386719

se	se/3	report:
0.0008	0.00026	106.3867
0.02	0.0066	106.387
0.06	0.02	106.39
9	3	106
50	17	110

Important points:

- Vocabulary
 - Standard deviation: variability in observations
 - Standard error: estimated precision of a statistic
- Which should you report?
 - What are you trying to describe: observations or statistics
 - And make sure you use the correct name
- How many digits should you report?
 - Kelley's rule: report to leading digit of $se/3$